Исходный код программы:

```
#include <fcntl.h>

#include <signal.h>

#include <stdio.h>

#include <stdlib.h>

#include <string.h>

#include <unistd.h>


int fd = -1;

volatile sig_atomic_t sigint_count = 0;


void sigHandler(int sig) {

  if (sig == SIGINT) {

    sigint_count++;

    if (sigint_count < 3) {

      const char msg[] = "\nПолучен сигнал SIGINT\n";

      write(fd, msg, sizeof(msg) - 1);

    } else {

      const char msg[] = "\nПолучен 3-й SIGINT. Завершение...\n";

      write(fd, msg, sizeof(msg) - 1);

      close(fd);

      exit(EXIT_SUCCESS);

    }

  } else if (sig == SIGQUIT) {

    const char msg[] = "\nПолучен сигнал SIGQUIT\n";

    write(fd, msg, sizeof(msg) - 1);

  }

}
```

```
int main() {
  fd = open("file.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);
  if (fd == -1) {
    perror("Error: open()");
    exit(EXIT_FAILURE);
  }
  signal(SIGINT, sigHandler);
  signal(SIGQUIT, sigHandler);

  int counter = 1;
  char buffer[64];

  while (1) {
    int len = sprintf(buffer, "%d\n", counter++);
    if (write(fd, buffer, len) == -1) {
      perror("Error: write()");
      close(fd);
      exit(EXIT_FAILURE);
    }
    sleep(1);
  }
  close(fd);
  exit(EXIT_SUCCESS);
}
```

Тест: два сигнала SIGINT, три сигнала SIGQUIT, сигнал SIGINT.

Сигнал SIGQUIT и SIGINT – вывод сообщения в файл, на третий SIGINT – завершение программы.

```
user@ubuntu-mate:~/eltex/module3/task5$ pgrep main
293530
user@ubuntu-mate:~/eltex/module3/task5$ kill -INT 293530
user@ubuntu-mate:~/eltex/module3/task5$ kill -INT 293530
user@ubuntu-mate:~/eltex/module3/task5$ kill -QUIT 293530
user@ubuntu-mate:~/eltex/module3/task5$ kill -QUIT 293530
user@ubuntu-mate:~/eltex/module3/task5$ kill -QUIT 293530
user@ubuntu-mate:~/eltex/module3/task5$ kill -INT 293530
user@ubuntu-mate:~/eltex/module3/task5$
```

```
user@ubuntu-mate:~/eltex/module3/task5$ make run
./main
user@ubuntu-mate:~/eltex/module3/task5$ cat file.txt
1
2
3
4
5
6
7
8
9
10
11
12

Получен сигнал SIGINT
13
14

Получен сигнал SIGINT
15
16
17
18
19
20
21

Получен сигнал SIGQUIT
22
23

Получен сигнал SIGQUIT
24

Получен сигнал SIGQUIT
25
26
27

Получен 3-й SIGINT. Завершение...
```

SIGABRT – завершение с дампом памяти.

```
user@ubuntu-mate:~/eltex/module3/task5$
user@ubuntu-mate:~/eltex/module3/task5$ pgrep main
306154
user@ubuntu-mate:~/eltex/module3/task5$ kill -ABRT 306154
user@ubuntu-mate:~/eltex/module3/task5$
```

```
user@ubuntu-mate:~/eltex/module3/task5$ make run
./main
make: *** [Makefile:7: run] Aborted (core dumped)
user@ubuntu-mate:~/eltex/module3/task5$ cat file.txt
1
2
3
4
5
6
7
8
9
10
11
12
13
14
user@ubuntu-mate:~/eltex/module3/task5$
```

SIGKILL- безусловное завершение.

```
user@ubuntu-mate:~/eltex/module3/task5$ make run
./main
make: *** [Makefile:7: run] Killed
user@ubuntu-mate:~/eltex/module3/task5$ cat file.txt
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
user@ubuntu-mate:~/eltex/module3/task5$
```

```
user@ubuntu-mate:~/eltex/module3/task5$
user@ubuntu-mate:~/eltex/module3/task5$ pgrep main
315624
user@ubuntu-mate:~/eltex/module3/task5$ kill -KILL 315624
user@ubuntu-mate:~/eltex/module3/task5$
```

SIGTERM – сигнал завершения.

```
user@ubuntu-mate:~/eltex/module3/task5$ pgrep main
317739
user@ubuntu-mate:~/eltex/module3/task5$ kill -TERM 317739
user@ubuntu-mate:~/eltex/module3/task5$
```

```
user@ubuntu-mate:~/eltex/module3/task5$ make run
./main
make: *** [Makefile:7: run] Terminated
user@ubuntu-mate:~/eltex/module3/task5$
```

SIGTSTP – сигнал остановки с терминала. SIGSTOP – сигнал остановки. SIGCONT – продолжить выполнение ранее остановленного процесса.

```
user@ubuntu-mate:~/eltex/module3/task5$ pgrep main
322359
user@ubuntu-mate:~/eltex/module3/task5$ kill -SIGTSTP 322359
user@ubuntu-mate:~/eltex/module3/task5$ pgrep main
322359
user@ubuntu-mate:~/eltex/module3/task5$ ps aux | grep main
user      218612  0.4  2.3 11841184 118004 ?     Sl   20:45   0:33 /home/user/.vscode-server/cli/se
rvers/Stable-41dd792b5e652393e7787322889ed5fdc58bd75b/server/node /home/user/.vscode-server/cli/ser
vers/Stable-41dd792b5e652393e7787322889ed5fdc58bd75b/server/out/server-main.js --connection-token=r
emotessh --accept-server-license-terms --start-server --enable-remote-auto-shutdown --socket-path=/
tmp/code-002d6a32-24a1-4327-a439-fff1359dbedd
user      322359  0.0  0.0   2644  1468 pts/0    T+   22:37   0:00 ./main
user      324426  0.0  0.0   9536  2648 pts/1    S+   22:39   0:00 grep --color=auto main
user@ubuntu-mate:~/eltex/module3/task5$
user@ubuntu-mate:~/eltex/module3/task5$ kill -SIGCONT 322359
user@ubuntu-mate:~/eltex/module3/task5$
```

```
user@ubuntu-mate:~/eltex/module3/task5$ make run
./main
^C^C^C^C^C^C^C^C^C
user@ubuntu-mate:~/eltex/module3/task5$ cat file.txt
1
2
3
4
5
6
7
8
9
10
11

Получен сигнал SIGINT
12
13
14

Получен сигнал SIGINT
15

Получен 3-й SIGINT. Завершение...
```

```
user@ubuntu-mate:~/eltex/module3/task5$ make run
./main
^C^C^C^C^C^C^C^C^C^C^C^C
user@ubuntu-mate:~/eltex/module3/task5$ cat file.txt
1
2
3
4
5
6
7
8
9
10

Получен сигнал SIGINT
11
12
13

Получен сигнал SIGINT
14

Получен 3-й SIGINT. Завершение...
user@ubuntu-mate:~/eltex/module3/task5$
```

```
user@ubuntu-mate:~/eltex/module3/task5$ pgrep main
328812
user@ubuntu-mate:~/eltex/module3/task5$ kill -SIGSTOP 328812
user@ubuntu-mate:~/eltex/module3/task5$ ps aux | grep main
user      218612  0.5  2.3 11841952 118964 ?     Sl   20:45   0:38 /home/user/.
vscode-server/cli/servers/Stable-41dd792b5e652393e7787322889ed5fdc58bd75b/serve
r/node /home/user/.vscode-server/cli/servers/Stable-41dd792b5e652393e7787322889
ed5fdc58bd75b/server/out/server-main.js --connection-token=remotessh --accept-s
erver-license-terms --start-server --enable-remote-auto-shutdown --socket-path=
/tmp/code-002d6a32-24a1-4327-a439-fff1359dbedd
user      328812  0.0  0.0   2644  1468 pts/0    T+   22:43   0:00 ./main
user      329223  0.0  0.0   9536  2652 pts/1    S+   22:44   0:00 grep --color
=auto main
user@ubuntu-mate:~/eltex/module3/task5$ kill -SIGCOUNT 328812
bash: kill: SIGCOUNT: invalid signal specification
user@ubuntu-mate:~/eltex/module3/task5$ kill -SIGCONT 328812
user@ubuntu-mate:~/eltex/module3/task5$
```